

OOPS Assignment 6

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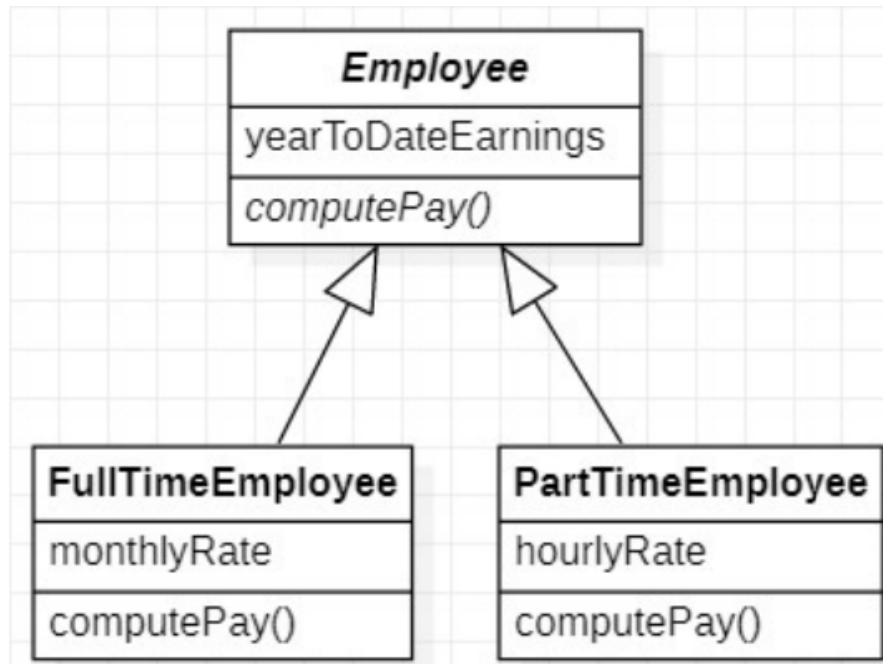
1. Try out all the code demonstrated in the lectures.

```
The decimal equivalent of 6C1 is 1729
12
In the outerclass k = 12
Hello this is the inner class

In the inner class k = 17
In the outerclass k = 17
Hello this is the inner class

Hello this is the inner class
22
The area of the rectangle is 120.0
The area of the triangle is 30.0
The strings are equal
ROI: 9.15
the pointer is a type of rectangle
The area is 20.0
```

2. We have discussed the following example as part of lecture on Abstract classes. Write a java program to implement the respective classes. (2 points)



```
The full time employee's pay after 2 months is 8000.0
Full time employee monthly rate 4000.0 and year to date earnings is 8000.0
The part time employee's pay after 100 hours is 10000.0
Part time employee monthly rate 100.0 and year to date earnings is 10000.0
```

3. Implement a class hierarchy, which has IMTechNumber at the top. Two classes derived from the IMTechNumber: IMTechInteger and IMTechComplex classes. (8 points)

```
The number is 36, its length is 2, it is a prime: false
The number is 13, its length is 2, it is a prime: true
false
1
After addition: The number is 49, its length is 2, it is a prime: false
After subtraction: The number is 23, its length is 2, it is a prime: true
After mutliplication: The number is 468, its length is 3, it is a prime: false
After division: The number is 2, its length is 1, it is a prime: true

-----COMPLEX NUMBERS-----

The number is 1 + 1i
The number is 1 + 2i
true
-1
After addition: The number is 2 + 3i
After subtraction: The number is 0 + -1i
After mutliplication: The number is -1 + 3i
After division: The number is 0 + 1i
Adding integer to complex number
Adding The number is 13, its length is 2, it is a prime: true to The number is 1 + 2i
The number is 14 + 2i
Adding The number is 36, its length is 2, it is a prime: false to The number is 1 + 1i
The number is 37 + 1i
```